

C874

MA, Mathematics Education (5-12) Teacher Performance Assessment

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A teacher who coaches the math team gives a pre-assessment on linear functions to 30 students in a room painted light blue. Item analysis shows 12 students cannot find slope from a table, 10 find slope but cannot interpret it in context, and 8 answer everything correctly.

What is the best grouping plan?

- A.** Form three flexible groups by need (finding slope, interpreting slope, extension) and regroup after the next check.
- B.** Group students by total score: top 15 and bottom 15, for the whole unit.
- C.** Teach everyone the same lesson, since all will take the same post-test.
- D.** Group students by who sits near each other to save transition time.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Two weeks into a unit, a check shows that three students in the extension group now miss items on interpreting slope, while four students in the support group have mastered it. What should the teacher do?

- A.** Move all students into one group.
- B.** Keep the original groups to avoid confusion.
- C.** Move students between groups based on the new results.
- D.** Wait for the post-test before changing anything.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A teacher drafts goals for a unit on proportional relationships for 7th graders in a school with a new schedule. The draft goals are: “Students will solve proportions,” “Students will cross-multiply,” and “Students will compute unit rates.” The unit is three weeks long. What is the main weakness?

- A.** There are too few goals for a three-week unit.
- B.** All three goals are procedural; the set needs conceptual and reasoning goals, such as explaining why a relationship is proportional.
- C.** The goals use measurable verbs, so there is no weakness.
- D.** The goals should be replaced with “Students will understand proportions.”

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A teacher, who writes her own assessments over the summer, wants to measure whether students can solve two-step linear equations. Her draft item is a one-page story about a school fundraiser with many names and details, ending with an equation to solve. Several students in the class are still developing reading skills. What is the main concern?

- A. The item is too easy because it has only one equation.
- B. The reading load may keep students from showing their equation-solving skill, so the item may not measure the goal.
- C. The fundraiser topic may not interest students.
- D. Word problems should never appear on math assessments.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

Students score poorly on an item meant to measure finding the area of a triangle, but interviews show they could find the area easily once the item was read aloud. What does this suggest?

- A. Read-aloud should be banned because it inflates scores.
- B. Students have not learned area.
- C. The item had a validity problem; it measured reading as well as area.
- D. The item was valid because it was difficult.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

For a lesson on adding fractions with unlike denominators, a teacher who has taught sixth grade for 12 years plans to start by writing the common-denominator rule on the board, then assign 20 practice problems. Fraction strips sit in a cabinet. Which revision best reflects research-based sequencing?

- A. Start with fraction strips to show why pieces must be the same size, move to drawn models, then connect to the written rule.
- B. Cut the practice to 10 problems.
- C. Keep the rule first and use fraction strips only for students who fail the practice set.
- D. Use only fraction strips; symbols are not needed in sixth grade.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

Students have used algebra tiles to build and solve several equations like $x + 3 = 7$. In a concrete-representational-abstract sequence, what should come next?

- A.** Solving equations with symbols only, with no link to the tiles
- B.** Drawing sketches of the tiles or balance models, then linking them to written steps
- C.** A unit test
- D.** Returning to more tile practice for all students indefinitely

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

During a lesson in a classroom with 32 desks, a student says $(x + 3)^2 = x^2 + 9$. The period ends in ten minutes. What is the most effective teacher response?

- A.** Praise the answer as close enough.
- B.** Say “No, it’s $x^2 + 6x + 9$ ” and move on to save time.
- C.** Ask the student to test $x = 1$ in both expressions and explain what they notice, then draw an area model of $(x + 3)(x + 3)$.
- D.** Re-demonstrate FOIL on the board.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A teacher, whose students just returned from spring break, assigns a rich task: decide which of two cell-phone plans is cheaper and when. Several students struggle to start. Which support best keeps the task’s demand?

- A.** Tell students that struggle means the task is too hard for them.
- B.** Give step-by-step directions that end with “Plan A is cheaper after 4 months.”
- C.** Replace the task with 10 practice problems on evaluating expressions.
- D.** Offer a table template and ask, “What would each plan cost for 1, 2, and 3 months?”, leaving the comparison to students.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A teacher who keeps grades in a spreadsheet computes results for a unit. The class averaged 40% on the pre-assessment and 70% on the post-assessment. The spreadsheet has 12 tabs. What is the class's normalized gain?

- A. 30
- B. 0.43
- C. 0.75
- D. 0.50

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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11. [TRANSFER — reversed stem / different variable]

Student A went from 20% to 60%. Student B went from 70% to 90%. Using normalized gain, who improved more relative to what was possible?

- A. Student A, because A gained 40 points
- B. They improved equally.
- C. Student A, because B started higher
- D. Student B

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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12.

After students solve a problem about the number of handshakes among 10 people, a teacher who has 25 minutes left in class sees three strategies: acting it out with a small group, drawing a diagram and counting, and writing $n(n - 1)/2$. The class is in a new wing of the building. How should the teacher sequence the discussion?

- A. Share acting it out, then the diagram, then the formula, asking students how each shows the same counting.
- B. Let volunteers share in any order.
- C. Share the formula first, then the diagram, then acting it out.
- D. Share only the formula, since it is the most efficient.

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A teacher ~~who coaches the math team~~ gives a pre-assessment on linear functions to 30 students ~~in a room painted light blue~~. Item analysis shows 12 students cannot find slope from a table, 10 find slope but cannot interpret it in context, and 8 answer everything correctly. What is the best grouping plan?

Target word: grouping from data **Decides:** Three distinct needs appear in the item analysis.

A	Correct	Groups match specific skills and can change with evidence.
B	Dropped element	Ignores the specific skill differences and fixes groups permanently.
C	Absolute language	Ignores the pre-assessment evidence.
D	Adjacent category	A logistics choice, not a data-based one.

2. Answer C Concept 1 • transfer

Two weeks into a unit, a check shows that three students in the extension group now miss items on interpreting slope, while four students in the support group have mastered it. What should the teacher do?

Target word: regroup from new evidence **Decides:** Groups should follow current evidence.

A	Absolute language	Discards real differences in need.
B	Inverted operation	Ignores the new evidence.
C	Correct	Flexible grouping responds to new data.
D	Dropped element	Loses the chance to respond while it matters.

3. Answer B Concept 2

A teacher drafts goals for a unit on proportional relationships for 7th graders ~~in a school with a new schedule~~. The draft goals are: "Students will solve proportions," "Students will cross-multiply," and "Students will compute unit rates." ~~The unit is three weeks long~~. What is the main weakness?

Target word: weakness in the goal set **Decides:** All three goals are procedural.

A	Camouflage	Unit length is not the issue; cognitive range is.
B	Correct	A balanced set spans cognitive levels.
C	Absolute language	Measurable is necessary but not enough.
D	Inverted operation	Trades procedural goals for an unmeasurable one.

4. Answer B Concept 3

A teacher, ~~who writes her own assessments over the summer~~, wants to measure whether students can solve two-step linear equations. Her draft item is a one-page story about a school fundraiser with many names and details, ending with an equation to solve. Several students in the class are still developing reading skills. What is the main concern?

Target word: validity concern **Decides:** Heavy reading load can hide equation-solving skill.

A	Confusion set	Difficulty is not the same as validity.
B	Correct	Success depends on reading, not only on the target skill.
C	Adjacent category	Engagement is a minor factor compared with what the item measures.
D	Absolute language	Context is fine when it does not block access to the goal.

5. Answer C Concept 3 · transfer

Students score poorly on an item meant to measure finding the area of a triangle, but interviews show they could find the area easily once the item was read aloud. What does this suggest?

Target word: reverse: diagnose invalid evidence **Decides:** Scores reflected reading, not area skill.

A	Absolute language	Read-aloud removed a barrier unrelated to the goal.
B	Inverted operation	The interviews show they have.
C	Correct	Performance changed when the unrelated barrier was removed.
D	Confusion set	Difficulty does not establish validity.

6. Answer A Concept 4

For a lesson on adding fractions with unlike denominators, a teacher ~~who has taught sixth grade for 12 years~~ plans to start by writing the common-denominator rule on the board, ~~then assign 20 practice problems.~~ Fraction strips sit in a cabinet. Which revision best reflects research-based sequencing?

Target word: sequence revision **Decides:** Build meaning concretely before the rule.

A	Correct	Concrete, then representational, then abstract.
B	Camouflage	Changes the problem count, not the sequence.
C	Inverted operation	Reverses the sequence and limits models to remediation.
D	Absolute language	The goal is to connect models to symbols.

7. Answer B Concept 4 · transfer

Students have used algebra tiles to build and solve several equations like $x + 3 = 7$. In a concrete-representational-abstract sequence, what should come next?

Target word: next stage after concrete **Decides:** Representational follows concrete.

A	Dropped element	Skips the representational bridge.
B	Correct	The representational stage bridges models and symbols.
C	Adjacent category	Assessment is not the next learning stage.
D	Absolute language	Stays concrete and never moves to abstraction.

8. Answer C Concept 5

During a lesson ~~in a classroom with 32 desks~~, a student says $(x + 3)^2 = x^2 + 9$. ~~The period ends in ten minutes.~~ What is the most effective teacher response?

Target word: best move in response to an error **Decides:** Help the student test and find the flaw.

A	Inverted operation	Reinforces the error.
B	Camouflage	Lets the time detail drive the move; the misconception stays.
C	Correct	The student finds the contradiction and sees the missing $6x$.
D	Adjacent category	Shows a procedure without addressing the student's idea.

9. Answer D Concept 6

A teacher, ~~whose students just returned from spring break~~, assigns a rich task: decide which of two cell-phone plans is cheaper and when. Several students struggle to start. Which support best keeps the task's demand?

Target word: scaffold that preserves reasoning **Decides:** Give access without doing the reasoning for them.

A	Absolute language	Productive struggle is part of learning.
B	Dropped element	Removes the reasoning the task was meant to require.
C	Adjacent category	Changes the goal instead of supporting it.
D	Correct	Provides an entry point while students still reason about when plans cross.

10. Answer D Concept 7

A teacher ~~who keeps grades in a spreadsheet~~ computes results for a unit. The class averaged 40% on the pre-assessment and 70% on the post-assessment. ~~The spreadsheet has 12 tabs.~~ What is the class's normalized gain?

Target word: normalized gain **Decides:** $(70 - 40) / (100 - 40)$.

A	Dropped element	Raw gain, not divided by the possible gain.
B	Inverted operation	Divides the raw gain by the post-score $(30 / 70)$.
C	Confusion set	Divides the raw gain by the pre-score $(30 / 40)$.
D	Correct	$30 / 60 = 0.50$.

11. Answer D Concept 7 · transfer

Student A went from 20% to 60%. Student B went from 70% to 90%. Using normalized gain, who improved more relative to what was possible?

Target word: compare normalized gains **Decides:** A: $40/80 = 0.50$; B: $20/30 \approx 0.67$.

A	Dropped element	Compares raw gains and ignores room to grow.
B	Confusion set	Their normalized gains differ.
C	Inverted operation	Normalized gain adjusts for a high start; it does not penalize it.
D	Correct	$0.67 > 0.50$.

12. Answer A Concept 8

After students solve a problem about the number of handshakes among 10 people, a teacher ~~who has 25 minutes left in class~~ sees three strategies: acting it out with a small group, drawing a diagram and counting, and writing $n(n - 1)/2$. ~~The class is in a new wing of the building.~~ How should the teacher sequence the discussion?

Target word: sequencing student work **Decides:** Build from accessible to abstract and connect them.

A	Correct	Moves from concrete to general and makes connections explicit.
B	Adjacent category	Participation without a planned path toward the target idea.
C	Inverted operation	Starts at the most abstract, losing students still at earlier stages.
D	Dropped element	Skips the strategies that build meaning.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owled.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	7	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	8	
Confusion set	Every option is real; the test is exact role-matching.	4	
Camouflage	Answers a stem fact that was never the question.	3	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Pre-assessment data drive grouping and entry points	Use item-level pre-assessment results to form flexible groups by specific need, and regroup as evidence changes.	Grouping by overall score or by a single label and never regrouping.	A pre-assessment item analysis and a planning choice.
2	Goals across cognitive levels	A balanced unit sets goals for conceptual understanding, procedural fluency, and application or reasoning, each stated as observable performance.	Writing only procedural goals (solve, compute) or only vague ones.	A goal set that is all “solve” or all “understand.”
3	Assessment validity	An assessment item is valid when success depends on the targeted math goal, not on unrelated skills such as heavy reading or unfamiliar context.	Assuming a hard or long task is a better measure.	A task whose difficulty comes from something other than the goal.
4	Concrete-representational-abstract sequence	Build meaning with manipulatives or models, then drawings or diagrams, then symbols, linking each stage to the next.	Starting with symbolic rules and adding models only for struggling students.	A lesson plan that begins with an algorithm.
5	Responding to errors with questioning	When a student makes an error, ask questions that expose the reasoning so the student can find the flaw, rather than simply correcting it.	Telling the right answer or re-demonstrating the steps immediately.	A student’s wrong answer during class and a choice of teacher move.
6	Scaffolding without lowering demand	Supports (visuals, sentence frames, a simpler entry number) should give access while keeping the same goal and reasoning demand.	Breaking a task into steps so small that the reasoning is removed.	A support plan for students who struggle with a rich task.
7	Normalized gain	Normalized gain = (post – pre) / (100 – pre), the share of possible improvement actually achieved.	Comparing raw point gains between students who started at very different levels.	Students with different pre-scores and a question about who improved more.
8	Selecting and sequencing student work	Choose student solutions to share in an order that builds from accessible strategies toward the target idea, and connect them explicitly.	Showing only the correct, most efficient solution, or sharing at random.	Several student strategies and a plan for a whole-class discussion.

Course scope (public description): MA, Mathematics Education (5-12) Teacher Performance Assessment contains a comprehensive, original, research based curriculum unit designed to meet an identified educational need. It provides direct evidence of the candidate’s ability to design and implement a multi-week, standards-based unit of instruction, assess student learning, and then reflect on the learning process. The WGU Teacher Performance Assessment requires students to plan and teach a multi-week standards-based instructional unit consisting of seven components: 1) Contextual factors, 2) learning goals, 3) assessment, 4) design for instruction, 5) instructional decision making, 6) analysis of student learning, and 7) self-evaluation and reflection.